

SCMission 2026

ROUND 2.2 CASE ANSWER

From Factory to Customer — Getting the Right Toys to the Right Place

TEAM NAME

LADYLOGI

PART A

Distribution & Logistics

PART B

Distributor Selection

PART C

AI / DIS Innovation

PART A - SCOPE & EXECUTIVE SUMMARY

Shipment planning: 32 shipment legs across 3 markets | 261,870 units | 596.57 CBM total

US MARKET	UK MARKET	AU MARKET
218,669 units	40,003 units	3,198 units
419.73 CBM \$1.06/unit dist. cost Burden vs ex-works: 1.19%	166.73 CBM \$1.18/unit dist. cost Burden vs ex-works: 1.68%	10.11 CBM \$1.72/unit dist. cost Burden vs ex-works: 5.74%

TOTAL UNITS	TOTAL CBM	TOTAL FREIGHT	TOTAL DIST. COST	PEAK GAP FLAGS	TARGET OFR
261,870	596.57	\$62,985.70	\$52852.5	5 shipments	≥98%

TOP 3 STRATEGIC RECOMMENDATIONS

HYBRID SHIPPING STRATEGY

Adopt a hybrid shipping strategy (FCL 40"/20" for large-volume shipments and LCL for small shipments) to achieve the best cost-service trade-off while securing container space during peak periods.

CONSOLIDATE US VOLUME INTO DC-US-W

Consolidate all U.S. volume into DC-US-W to optimize cost efficiency and operational simplicity. The facility offers sufficient storage capacity (500 CBM) to accommodate total inbound volume, while its DC handling cost is only marginally lower (by USD 18/CBM) compared to DC-US-E , making it the more economical choice overall. Additionally, routing shipments via the Trans-Pacific lane to the West Coast ensures a shorter, more direct transit, reducing lead time variability and minimizing exposure to geopolitical risks. This approach prioritizes speed, cost savings, and supply chain reliability over splitting volume across multiple distribution centers.

REDUCE AU LAST-MILE BURDEN

Focus cost-reduction efforts on the Australian market, which shows the highest distribution burden (5.74%). Negotiate better last-mile rates or consolidate multi-unit orders to bring the burden closer to the US/UK levels.

A1 - SHIPMENT PIPELINE MASTER TABLE | DC Allocation & Peak Readiness

32 shipment legs | ETD: May 15 – Jul 31, 2026 | Destination DC: DC-US-W / DC-UK / DC-AU

DC ALLOCATION LOGIC

- US (SKUs A, B, F) → DC-US-W (Los Angeles, CA)
 - ✓ \$18/CBM handling (vs \$20 at DC-US-E)
 - ✓ 500 CBM capacity, 50-day ocean LT
 - ✓ Trans-Pacific route = shorter, lower risk
- UK (SKUs C, D) → DC-UK (Birmingham, UK)
 - ✓ 40-day ocean LT, \$22/CBM handling, 300 CBM
- AU (SKU E) → DC-AU (Sydney, AU)
 - ✓ 25-day ocean LT, \$25/CBM handling, 200 CBM



PEAK WINDOW & READINESS LOGIC

- Gap (days) = Relevant Peak Start – Expected DC Arrival
- READY BEFORE PEAK: Gap ≥ 0 (arrives before peak)
- GAP - ACTION NEEDED: Gap < 0 (arrives after peak)
 - Action: Switch to air freight/split shipments
 - Air transit times (Vina Dowell reference):
 - US: 6–14 days · UK: 5–7 days · AU: 5–8 days
- Peak periods: US = Oct–Dec / Aug | UK = Nov–Jun
AU = Nov–Dec / Jan & Jul (school hols)

PEAK READINESS STATUS SUMMARY

READY BEFORE PEAK

**27 shipments
(84%)**

All 32 legs arrive before their respective peak window

GAP — ACTION NEEDED

**5 shipments
(16%)**

US: IDs 15,18,20,23,26
Split shipments or switch to air freight

Air Freight Option

**6-14 days US
5-7 days UK / 5-8 days AU**

Switch time-critical shipments to air to guarantee ≥98% service level

DC-US-W Capacity

**419.73 CBM used
500 CBM total (84%)**

Within safe utilization range
No overflow risk to DC-US-E

ID	ETD	DC	Market	SKU Mix (abbrev.)	Units	CBM	Mode	Dist. Cost	Readiness
3	15-May	DC-US-W	US West	A:20k, B:5,502, F:4,200	29,702	62.76	FCL40	\$30,091	✅ READY
6	22-May	DC-US-W	US West	A:20k, B:10,116, F:3,802	33,918	69.64	FCL40+LCL	\$34,516	✅ READY
8	29-May	DC-US-W	US West	A:11k, B:3,323, F:4,500	18,823	37.12	FCL20+LCL	\$20,551	✅ READY
7	29-May	DC-UK	UK	C:1,816, D:1,790	3,606	11.9	LCL	\$3,980	✅ READY
15	19-Jun	DC-US-W	US West	A:7,058, B:5,309, F:4,500	16,867	30.39	FCL20+LCL	\$17,519	⚠️ GAP
18	26-Jun	DC-US-W	US West	A:2,040, B:5,309, F:2,200	9,549	15.61	FCL20	\$10,920	⚠️ GAP
20	03-Jul	DC-US-W	US West	A:3,541, B:3,000, F:5,000	11,541	18.45	FCL20	\$12,565	⚠️ GAP
21	10-Jul	DC-AU	AU	E:700	700	2.21	LCL	\$973	✅ READY
23	10-Jul	DC-US-W	US West	A:12,041, B:3,000, F:4,300	19,341	39.02	FCL40	\$21,375	⚠️ GAP
25	17-Jul	DC-UK	UK	C:2,500, D:500	3,000	13.29	LCL	\$3,622	✅ READY

**27/32 shipments
arrive READY BEFORE
PEAK · → 5 flagged
gaps require ETD pull-
forward or air freight
escalation**

A2 - SHIPPING MODE SELECTION | Decision Framework & Rate Card

Cost-optimal mode selected for each of 32 shipment legs | Policy: minimize total freight cost while securing peak readiness

ZONE 1: CBM < 15	ZONE 2: 15 ≤ CBM ≤ 28	ZONE 3: 28 < CBM ≤ 65	ZONE 4: CBM > 65
Default to LCL Quick-check vs 20ft FCL rate if competitive	Compare LCL total vs 20ft FCL flat rate Select lower cost	Compare: All-LCL vs 40ft FCL vs 20ft FCL + overflow LCL	40ft FCL as base + LCL for overflow volume only
→ LCL	→ LCL or FCL 20ft	→ Best of 3 options	→ FCL 40ft + LCL

Additional rule: If gap vs peak start < 0 → Flag for air freight review while keeping ocean cost as baseline

FREIGHT RATE CARD (Case Inputs)									
Market	FCL 20ft Rate	FCL 40ft Rate	LCL Rate/CBM	20ft Capacity	40ft Capacity	FCL Lead Time	LCL Lead Time	DC Handling	Last-mile/Unit
US (→DC-US-W)	\$3,000	\$5,200	\$200/CBM	28 CBM	65 CBM	50 days	55 days	\$18/CBM	\$0.80
UK (→DC-UK)	\$2,500	\$4,200	\$70/CBM	28 CBM	65 CBM	40 days	45 days	\$22/CBM	\$0.80
AU (→DC-AU)	\$1,100	\$2,000	\$35/CBM	28 CBM	65 CBM	25 days	30 days	\$25/CBM	\$1.20

DECISION EXAMPLE: Shipment #6 (ETD May 22) — A:20k + B:10,116 + F:3,802 = 33,918 units = 69.64 CBM → Exceeds 400ft (65 CBM); compare 40ft+overflow vs all-LCL

All LCL	69.64 × \$200 = \$13,928	Highest cost; no container risk	✗
FCL 20ft + Overflow LCL	\$3,000 + (69.64–28)×\$200 = \$11,328	Partial control; mid cost	✗
FCL40 + Overflow LCL Cost [SELECTED]	\$5,200 + (69.64–65)×\$200 = \$6,128	Lowest cost; full container control; \$6,128 with overflow LCL	✓

Policy applied consistently across all 32 shipments · All A1 cost checks confirmed \$0 variance between A1 and A2

A2 — MODE DECISION RESULTS | 32 Shipments Across All Markets

All modes selected by lowest total freight cost · 5 timing-gap shipments flagged for air escalation · 0 cost discrepancies vs A1 baseline

Shipping Mode	# Shipments	Units	CBM	Total Freight	Avg Freight/CBM	Market	Shipments	Units	CBM	Freight Cost	\$/Unit
LCL	21	57,347	193.21	\$15,455.70	\$79.99	US (DC-US-W)	12	218,669	419.73	\$49,894.00	\$0.228
FCL 20ft	3	21,390	35.01	\$7,100.00	\$202.80	UK (DC-UK)	12	40,003	166.73	\$11,671.10	\$0.292
FCL 20ft + LCL	2	35,690	67.51	\$8,302.00	\$122.97	AU (DC-AU)	8	3,198	10.11	\$1,420.60	\$0.444
FCL 40ft	5	113,525	231.20	\$26,000.00	\$112.46						

FULL SHIPMENT MODE DECISION TABLE (All 32 Legs)										
ID	ETD	DC	CBM	Rec. Mode	Rec. Freight	Alt. Mode	Alt. Cost	Savings vs Alt.	Gap(d)	Status
1	15-May	DC-AU	0.95	LCL	\$33.25	FCL 20ft / 20ft+LCL	\$1,100	+\$1,067	17	✅ READY
2	15-May	DC-UK	1.78	LCL	\$124.60	FCL 20ft / 20ft+LCL	\$2,500	+\$2,375	125	✅ READY
3	15-May	DC-US-W	62.76	FCL 40ft	\$5,200	FCL 20ft / 20ft+LCL	\$9,952	+\$4,752	28	✅ READY
4	22-May	DC-AU	2.2	LCL	\$77.00	FCL 20ft / 20ft+LCL	\$1,100	+\$1,023	10	✅ READY
5	22-May	DC-UK	1.78	LCL	\$124.60	FCL 20ft / 20ft+LCL	\$2,500	+\$2,375	118	✅ READY
6	22-May	DC-US-W	69.64	FCL 40ft+LCL	\$6,128	FCL 20ft / 20ft+LCL	\$11,328	+\$5,200	16	✅ READY
7	29-May	DC-UK	11.9	LCL	\$833.00	FCL 40ft / 40ft+LCL	\$2,500	+\$1,667	111	✅ READY
8	29-May	DC-US-W	37.12	FCL 20ft+LCL	\$4,824	LCL	\$5,200	+\$376	14	✅ READY
9	05-Jun	DC-AU	0.95	FCL 20ft*	\$1,100	FCL 20ft / 20ft+LCL	\$33	-\$1,067	1	✅ READY*
10	05-Jun	DC-UK	4.02	LCL	\$281.40	FCL 20ft / 20ft+LCL	\$2,500	+\$2,219	104	✅ READY
11	05-Jun	DC-US-W	39.97	FCL 40ft	\$5,200	FCL 20ft / 20ft+LCL	\$5,394	+\$194	7	✅ READY
12	12-Jun	DC-UK	3.85	LCL	\$269.50	FCL 20ft / 20ft+LCL	\$2,500	+\$2,231	97	✅ READY
13	12-Jun	DC-US-W	50.04	FCL 40ft	\$5,200	FCL 20ft / 20ft+LCL	\$7,408	+\$2,208	0	✅ READY
14	19-Jun	DC-UK	22.68	LCL	\$1,587.60	FCL 40ft / 40ft+LCL	\$2,500	+\$912	90	✅ READY
15	19-Jun	DC-US-W	30.39	FCL 20ft+LCL	\$3,478	FCL 20ft / 20ft+LCL	\$5,200	+\$1,722	-12	⚠️ GAP
16	26-Jun	DC-AU	0.95	LCL	\$33.25	FCL 20ft / 20ft+LCL	\$1,100	+\$1,067	98	✅ READY
17	26-Jun	DC-UK	36.59	LCL	\$2,561.30	FCL 20ft / 20ft+LCL	\$3,101	+\$540	83	✅ READY
18	26-Jun	DC-US-W	15.61	FCL 20ft	\$3,000	LCL	\$3,122	+\$122	-14	⚠️ GAP
19	03-Jul	DC-UK	39.26	LCL	\$2,748.20	FCL 20ft / 20ft+LCL	\$3,288	+\$540	76	✅ READY
20	03-Jul	DC-US-W	18.45	FCL 20ft	\$3,000	LCL	\$3,690	+\$690	-21	⚠️ GAP

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All modes selected by lowest total freight cost · 5 timing-gap shipments flagged for air escalation · 0 cost discrepancies vs A1 baseline

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Market	Shipments	Units	CBM	Freight Cost	\$/Unit
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FULL SHIPMENT MODE DECISION TABLE (All 32 Legs)										
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21	10-Jul	DC-AU	2.21	LCL	\$77.35	FCL 20ft	\$1,100	+\$1,023	84	✅ READY
22	10-Jul	DC-UK	13.29	LCL	\$930.30	FCL 20ft	\$2,500	+\$1,570	69	✅ READY
23	10-Jul	DC-US-W	39.02	FCL 40ft	\$5,200	FCL 20ft+LCL	\$5,204	+\$4	-28	⚠️ GAP
24	17-Jul	DC-AU	0.95	LCL	\$33.25	FCL 20ft	\$1,100	+\$1,067	77	✅ READY
25	17-Jul	DC-UK	13.29	LCL	\$930.30	FCL 20ft	\$2,500	+\$1,570	62	✅ READY
26	17-Jul	DC-US-W	39.41	FCL 40ft	\$5,200	FCL 20ft+LCL	\$5,282	+\$82	-35	⚠️ GAP
27	24-Jul	DC-AU	0.95	LCL	\$33.25	FCL 20ft	\$1,100	+\$1,067	70	✅ READY
28	24-Jul	DC-UK	13.29	LCL	\$930.30	FCL 20ft	\$2,500	+\$1,570	55	✅ READY
29	24-Jul	DC-US-W	9.35	LCL	\$1,870	FCL 20ft	\$3,000	+\$1,130	14	✅ READY
30	31-Jul	DC-AU	0.95	LCL	\$33.25	FCL 20ft	\$1,100	+\$1,067	63	✅ READY
31	31-Jul	DC-UK	5.0	LCL	\$350.00	FCL 20ft	\$2,500	+\$2,150	48	✅ READY
32	31-Jul	DC-US-W	7.97	LCL	\$1,594	FCL 20ft	\$3,000	+\$1,406	7	✅ READY

A3 — TOTAL LANDED COST & DISTRIBUTION BURDEN ANALYSIS

Formula: Ocean Freight + DC Handling + Last-Mile = Distribution Cost | Burden = Distribution Cost ÷ Ex-Works Value

Market	Units	CBM	Ocean Freight	DC Handling	Last-Mile Cost	Total Dist. Cost	Dist. Cost/Unit	Ex-Works/Unit	Full Landed/Unit	Burden %	Flag
US	218,669	419.73	\$49,894.00	\$7,555.14	\$174,935.20	\$232,384.34	\$1.063	\$89.33	\$90.40	1.19%	—
UK	40,003	166.73	\$11,671.10	\$3,668.06	\$32,002.40	\$47,341.56	\$1.183	\$70.28	\$71.46	1.68%	—
AU	3,198	10.11	\$1,420.60	\$252.75	\$3,837.60	\$5,510.95	\$1.723	\$30.00	\$31.72	5.74%	⚠️ HIGHEST
TOTAL	261,870	596.57	\$62,985.70	\$11,475.95	\$210,775.20	\$285,236.85	\$1.089	—	—	—	

COST COMPONENT SHARE BY MARKET (% of total distribution cost)					
Market	Ocean Share	DC Handling Share	Last-Mile Share	Primary Cost Driver	Action Priority
US	21.47%	3.25%	75.28%	Last-Mile (\$0.80/unit)	NORMAL — Monitor rate creep
UK	24.65%	7.75%	67.60%	Last-Mile (\$0.80/unit)	NORMAL — Monitor rate creep
AU	25.78%	4.59%	69.64%	Last-Mile (\$1.20/unit)	⚠️ HIGH — Reduce last-mile cost

🔍 AUSTRALIA: HIGHEST BURDEN ANALYSIS & IMPROVEMENT PLAN

WHY HIGHEST BURDEN?

- SKU E (Scooter) ex-works: \$30/unit
- Last-mile: \$1.20/unit = 4.0% alone
- Last-mile = 69.6% of all AU dist. cost
- Total burden: 5.74% vs 1.19% US

IMPROVEMENT ACTION

- Consolidate multi-unit AU orders
- Renegotiate parcel rate with 3PL
- Target: reduce to \$1.00/unit (-17%)
- Engage DownUnder Toys immediately

FINANCIAL IMPACT

- Current last-mile: \$3,837.60
- Target last-mile: \$3,198.00
- Savings: \$639.60 (~16.7% reduction)
- AU burden drops: 5.74% → ~5.08%

NEXT STEPS

- Initiate rate negotiation Q3 2026
- Set volume commitment threshold
- Review bundling policy with DC-AU
- Track monthly AU burden vs target

Last-mile accounts for 75.3% (US), 67.6% (UK), 69.6% (AU) of distribution cost · AU carries 5.74% burden vs 1.19% US — highest improvement ROI

A3 — SHIPMENT-LEVEL LANDED COST DETAIL | All 32 Shipments

Distribution Cost = Ocean Freight + DC Handling Cost + Last-Mile Cost · Burden = Dist. Cost / Ex-Works Value

ID	ETD	DC	Units	CBM	Ocean Freight	DC Handling	Last-Mile	Dist. Cost	Cost/Unit	Ex-Works/Unit	Burden %	Cost Driver
1	15-May	DC-AU	300	0.95	\$33.25	\$23.75	\$360.00	\$417.00	\$1.39	\$30.00	4.63%	Last-Mile
2	15-May	DC-UK	540	1.78	\$124.60	\$39.16	\$432.00	\$595.76	\$1.10	\$60.00	1.84%	Last-Mile
3	15-May	DC-US-W	29,702	62.76	\$5,200.00	\$1,129.68	\$23,761.60	\$30,091.28	\$1.01	\$93.62	1.08%	Last-Mile
4	22-May	DC-AU	698	2.2	\$77.00	\$55.00	\$837.60	\$969.60	\$1.39	\$30.00	4.63%	Last-Mile
5	22-May	DC-UK	540	1.78	\$124.60	\$39.16	\$432.00	\$595.76	\$1.10	\$60.00	1.84%	Last-Mile
6	22-May	DC-US-W	33,918	69.64	\$6,128.00	\$1,253.52	\$27,134.40	\$34,515.92	\$1.02	\$88.41	1.15%	Last-Mile
7	29-May	DC-UK	3,606	11.9	\$833.00	\$261.80	\$2,884.80	\$3,979.60	\$1.10	\$60.14	1.83%	Last-Mile
8	29-May	DC-US-W	18,823	37.12	\$4,824.00	\$668.16	\$15,058.40	\$20,550.56	\$1.09	\$92.59	1.18%	Last-Mile
9	05-Jun	DC-AU	300	0.95	\$1,100.00	\$23.75	\$360.00	\$1,483.75	\$4.95	\$30.00	16.49%	Ocean*
10	05-Jun	DC-UK	804	4.02	\$281.40	\$88.44	\$643.20	\$1,013.04	\$1.26	\$80.00	1.58%	Last-Mile
11	05-Jun	DC-US-W	19,346	39.97	\$5,200.00	\$719.46	\$15,476.80	\$21,396.26	\$1.11	\$96.20	1.15%	Last-Mile
12	12-Jun	DC-UK	1,271	3.85	\$269.50	\$84.70	\$1,016.80	\$1,371.00	\$1.08	\$56.99	1.89%	Last-Mile
13	12-Jun	DC-US-W	25,396	50.04	\$5,200.00	\$900.72	\$20,316.80	\$26,417.52	\$1.04	\$88.32	1.18%	Last-Mile
14	19-Jun	DC-UK	5,673	22.68	\$1,587.60	\$498.96	\$4,538.40	\$6,624.96	\$1.17	\$68.29	1.71%	Last-Mile
15	19-Jun	DC-US-W	16,867	30.39	\$3,478.00	\$547.02	\$13,493.60	\$17,518.62	\$1.04	\$85.26	1.22%	Last-Mile
16	26-Jun	DC-AU	300	0.95	\$33.25	\$23.75	\$360.00	\$417.00	\$1.39	\$30.00	4.63%	Last-Mile
17	26-Jun	DC-UK	9,374	36.59	\$2,561.30	\$804.98	\$7,499.20	\$10,865.48	\$1.16	\$67.19	1.73%	Last-Mile
18	26-Jun	DC-US-W	9,549	15.61	\$3,000.00	\$280.98	\$7,639.20	\$10,920.18	\$1.14	\$73.75	1.55%	Last-Mile
19	03-Jul	DC-UK	8,195	39.26	\$2,748.20	\$863.72	\$6,556.00	\$10,167.92	\$1.24	\$77.56	1.60%	Last-Mile
20	03-Jul	DC-US-W	11,541	18.45	\$3,000.00	\$332.10	\$9,232.80	\$12,564.90	\$1.09	\$85.50	1.27%	Last-Mile
21	10-Jul	DC-AU	700	2.21	\$77.35	\$55.25	\$840.00	\$972.60	\$1.39	\$30.00	4.63%	Last-Mile
22	10-Jul	DC-UK	3,000	13.29	\$930.30	\$292.38	\$2,400.00	\$3,622.68	\$1.21	\$73.33	1.65%	Last-Mile
23	10-Jul	DC-US-W	19,341	39.02	\$5,200.00	\$702.36	\$15,472.80	\$21,375.16	\$1.11	\$93.91	1.18%	Last-Mile
24	17-Jul	DC-AU	300	0.95	\$33.25	\$23.75	\$360.00	\$417.00	\$1.39	\$30.00	4.63%	Last-Mile
25	17-Jul	DC-UK	3,000	13.29	\$930.30	\$292.38	\$2,400.00	\$3,622.68	\$1.21	\$73.33	1.65%	Last-Mile
26	17-Jul	DC-US-W	19,740	39.41	\$5,200.00	\$709.38	\$15,792.00	\$21,701.38	\$1.10	\$93.83	1.17%	Last-Mile
27	24-Jul	DC-AU	300	0.95	\$33.25	\$23.75	\$360.00	\$417.00	\$1.39	\$30.00	4.63%	Last-Mile
28	24-Jul	DC-UK	3,000	13.29	\$930.30	\$292.38	\$2,400.00	\$3,622.68	\$1.21	\$73.33	1.65%	Last-Mile
29	24-Jul	DC-US-W	7,746	9.35	\$1,870.00	\$168.30	\$6,196.80	\$8,235.10	\$1.06	\$76.44	1.39%	Last-Mile
30	31-Jul	DC-AU	300	0.95	\$33.25	\$23.75	\$360.00	\$417.00	\$1.39	\$30.00	4.63%	Last-Mile
31	31-Jul	DC-UK	1,000	5.0	\$350.00	\$110.00	\$800.00	\$1,260.00	\$1.26	\$80.00	1.58%	Last-Mile
32	31-Jul	DC-US-W	6,700	7.97	\$1,594.00	\$143.46	\$5,360.00	\$7,097.46	\$1.06	\$77.46	1.37%	Last-Mile

* Shipment 9: FCL 20ft used for timing reasons despite LCL being cheaper — ensures peak readiness · Red rows = AU burden ≥4.63% · Dist. cost driver = Last-Mile in all 32 cases

PART B: Distributor Selection

SELECTION FRAMEWORK

MK evaluates each distributor using a structured 4-question model aligned with supply chain performance and market behavior data.

C1

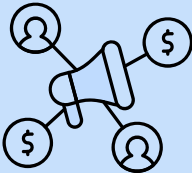


Warehouse Capacity

Weight: 25%

Distributor warehouse CBM vs. peak weekly DC replenishment volume. Risk of overflow during surge weeks?

C2



Channel Fit

Weight: 30%

Does the distributor reach the dominant purchase channels in each market e-commerce, premium retail, or indie stores?

C3



Order Fulfillment Rate

Weight: 25%

Order Fulfillment Rate vs. MK's ≥98% SLA target. Distributors failing this threshold are disqualified as primary partners.

C4



Contingency

Weight: 20%

If OFR drops to 88% for 4 peak weeks: quantify revenue at risk, assess fallback options, and define MK's protection plan.

Scoring scale: 1 = Poor | 2 = Below Average | 3 = Average | 4 = Good | 5 = Excellent | Weighted Score determines final recommendation per market

DISTRIBUTOR SCORECARD

Five weighted criteria evaluated across all candidates — selected partners highlighted in each market

Distributor	Mkt	Rev (USD M)	WH Cap (CBM)	OFR (%)	C1 Vol (25%)	C2 Channel (30%)	C3 OFR (25%)	Weighted Score	Status
PlayGiant USA	US	\$120M	800	97.8%	5	5	4	4.55	✓ SELECTED
KidWorld Distrib.	US	\$88M	600	96.3%	4	3	3	3.45	✗ Not Selected
BrightToy UK	UK	\$42M	280	98.2%	4	4	5	4.15	✓ SELECTED
JoyZone UK	UK	\$28M	180	91.5%	3	2	1	2.00	✗ Disqualified
DownUnder Toys	AU	\$31M	220	96.8%	5	4	4	4.25	✓ SELECTED

B1: CAN THEY HANDLE THE VOLUME?

Distributor warehouse CBM vs. peak weekly DC replenishment — assessing overflow risk across all three markets

US Market

PlayGiant USA

DC Allocated	DC-US-W + DC-US-E
Products	Products A, B, F
Peak Weekly CBM	69.6 CBM
Peak Week	May 22, 2026
Total May–Jul CBM	415.2 CBM
WH Capacity	800 CBM
WH Utilization	8.7%
Buffer vs. Peak	11.5x peak week

✓ WELL WITHIN CAPACITY

UK Market

BrightToy UK

DC Allocated	DC-UK (Birmingham)
Products	Products C, D
Peak Weekly CBM	39.26 CBM
Peak Week	Jul 3, 2026
Total May–Jul CBM	166.71 CBM
WH Capacity	280 CBM
WH Utilization	14.0%
Buffer vs. Peak	7.1x peak week

✓ WELL WITHIN CAPACITY

AU Market

DownUnder Toys

DC Allocated	DC-AU (Sydney)
Products	Product E
Peak Weekly CBM	2.21 CBM
Peak Week	Jul 10, 2026
Total May–Jul CBM	10.07 CBM
WH Capacity	220 CBM
WH Utilization	1.0%
Buffer vs. Peak	99.8x peak week

✓ MASSIVELY OVER-SPEC

Key Insight: All three selected distributors have ample warehouse capacity - peak utilization never exceeds 14% of warehouse CBM, with PlayGiant USA providing the largest absolute buffer at 730 CBM.

B2: CHANNEL FIT

Does the distributor reach the right customers?

US — PlayGiant USA

Q2: Channel

- Online-first 62% (Amazon, Walmart.com) + 38% B&M
- ✓ Strong e-commerce → direct Amazon/Walmart.com integration
- △ Gap: Limited Midwest B&M coverage (~15% US retail gap)

Score: 4.55 / 5.0 ✓ EXCELLENT

UK — BrightToy UK

Q2: Channel

- Mixed: 48% online + 32% indie toy shops + 20% other
- ✓ Premium retail partner; serves indie shops; high ESG standards
- △ Gap: ESG audit required: EN71 + REACH compliance before first shipment

Score: 4.15 / 5.0 ✓ GOOD FIT

AU — DownUnder Toys

Q2: Channel

- Online 55% + In-store 45%; Safety + AU safety mark key drivers
- ✓ Only national AU distributor; NSW + VIC = 60% of AU population
- △ Gap: No backup distributor in AU market (single-source dependency)

Score: 4.25 / 5.0 ✓ ADEQUATE

B3: KEY RISK PER MARKET

What is the one identified risk - with proposed mitigation?

US — PlayGiant USA

Q3: Key Risk

CHANNEL GAP [MEDIUM]

PlayGiant lacks Midwest brick-and-mortar coverage → ~15% B&M market missed

- Mitigation: Co-activate KidWorld for Midwest B&M + add KPI reporting clause (Q2 2027)

Timeline: 90 days

UK — BrightToy UK

Q3: Key Risk

ESG COMPLIANCE [HIGH]

BrightToy requires MK to meet EN71 & REACH before distribution — non-compliance risks contract termination

- Mitigation: Pre-audit MK products (certified lab). Budget \$15K–\$25K. Build ESG product passport. Complete within 60 days.

Timeline: 60 days

AU — DownUnder Toys

Q3: Key Risk

SINGLE-SOURCE RISK [CRITICAL]

No backup distributor in AU — if DownUnder fails, MK has zero national AU market reach

- Mitigation: 30-day SLA notice + penalty clause. Pre-identify 3PLs (Australia Post Business, Toll Group). Emergency Shopify AU D2C channel (12-wk setup).

Timeline: 120 days

All identified risks are manageable within 60–120 days. BrightToy UK's ESG compliance requirement is the highest-urgency action item — must be initiated before first shipment.

MARKET RECOMMENDATIONS

One distributor selected per market based on weighted scoring — rationale structured around the 4-question framework

US Market

vs. KidWorld Distrib. — 3.45

- 800 CBM warehouse — 11.5x peak weekly volume (69.6 CBM)
- E-commerce strength aligns with US 62% online purchase behavior
- Largest revenue scale (\$120M) — lowest financial risk to MK
- 97.8% OFR — closest to 98% target among US candidates

→ KidWorld retained as regional Midwest B&M partner (co-activation plan)

UK Market

vs. JoyZone UK — 2.00 (DISQUALIFIED)

- ONLY distributor meeting MK's ≥98% SLA requirement (98.2%)
- Serves premium retail + indie toy shops (UK's key 32% channel)
- ESG-premium positioning matches UK's eco-credential purchase driver
- 280 CBM warehouse — 7.1x peak weekly UK volume (39.26 CBM)

→ JoyZone disqualified: 91.5% OFR = 6.5pp below MK's minimum threshold

AU Market

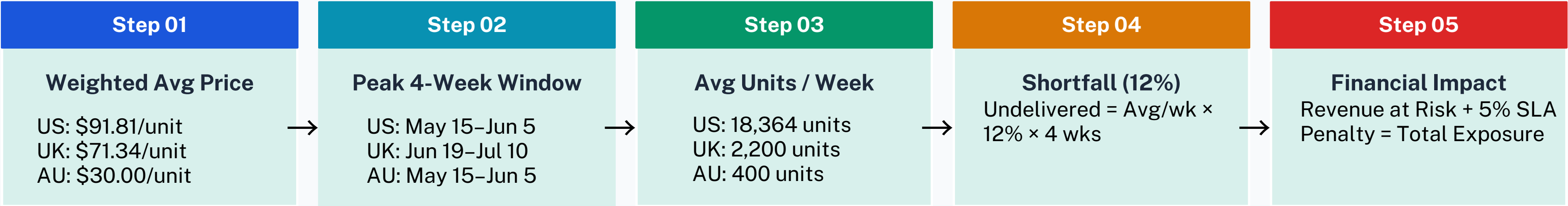
vs. No listed alternative

- Only national Australian distributor — default strategic selection
- NSW + VIC coverage = 60% of AU population; aligned with DC-AU Sydney
- 220 CBM warehouse — 99.8x peak weekly AU volume (2.21 CBM)
- Mixed online/in-store distribution matches AU 55%/45% behavior

→ Emergency 3PL backstop required: zero fallback option in AU market

B4: CONTINGENCY ANALYSIS - 88% OFR FOR 4 PEAK WEEKS

Scenario: Primary distributor fulfillment drops to 88% over 4 consecutive peak weeks. Quantify revenue impact and define MK's fallback action plan.



Step 1 Detail: Weighted Average Ex-Works Price Calculation (W5–W16: May 15 – Jul 31, 2026)

Market	Product	Ex-Works Price	Volume (W5–W16)	Revenue Contribution	Weighted Avg Price
US	A (\$105) + B (\$55) + F (\$90)	\$105 / \$55 / \$90	121,987 / 41,487 / 44,720	\$12.81M + \$2.28M + \$4.02M	\$91.81 / unit
UK	C (\$80) + D (\$40)	\$80 / \$40	31,119 / 8,594	\$2.49M + \$0.34M	\$71.34 / unit
AU	E (\$30) only	\$30	3,400	\$102,000	\$30.00 / unit

TOP 3 STRATEGY FOR MK

- 1.Add SLA triggers: OFR <95% for 4 weeks → fallback or direct-ship in 14 days.
- 2.Launch BrightToy UK EN71 + REACH audit now to prevent premium retail delisting.
- 3.Pre-sign AU 3PL backup (DHL / Toll) for 72-hour activation.

ACTION PLAN AND STRATEGIC RECOMMENDATION

Market	Fallback Distributor	Switch Triggered?	Action Plan
US - PlayGiant USA (97.8%)	KidWorld Distrib. (96.3% OFR)	✓ SHOULD Switch	▶ Pre-negotiate standby agreement with KidWorld immediately ▶ Trigger: OFR < 95% over any rolling 4-week period ▶ Bridge gap using 6-week safety stock at DC-US ▶ Post-transition: KidWorld handles Midwest B&M; PlayGiant retains e-commerce
UK - BrightToy UK (98.2%)	JoyZone UK (91.5% OFR)	✗ SHOULD NOT Switch to JoyZone	▶ JoyZone's 91.5% OFR is itself below MK's 98% threshold — do NOT switch as primary ▶ Activate 2-week buffer stock at DC-UK + emergency 3PL (DHL/XPO UK) ▶ 3PL cost: ~\$8/CBM × 166.7 CBM = ~\$1,333 during disruption ▶ Enforce SLA penalty: £500/unit for violations exceeding 2% shortfall
AU DownUnder Toys (96.8%)	No listed alternative (Sole national option)	⊘ NO FALLBACK Preventive only	▶ 6-week safety stock at DC-AU (200 CBM >> 2.19 CBM peak = fully sufficient) ▶ SLA clause: AUD \$80/unit penalty + 30-day notice period required ▶ Contingency: Activate emergency Shopify AU D2C if OFR < 90% for 2 consecutive weeks ▶ Long-term: Qualify secondary regional 3PL in WA (Australia Post, Toll Group)

DECISION LOGIC

Decision Order	IF Condition	THEN Action	Explanation / Formula
1	Destination is DC-US-E or DC-UK and port congestion is triggered	Recalculate ETA and ocean cost, then send shipment to recovery assessment	New ETA = original ETA + 15 days; Ocean cost = base × 1.20
2	Order is Product A and destination/due date is DC-US-E May 15 or DC-UK Jun 12	Tag as highest priority and evaluate before other products	These are case-defined critical orders
3	Both priority Product A orders have delay days > 0	Protect 20,000 pcs DC-US-E first, then 13,773 pcs DC-UK	Priority = earliest due date (May 15 < Jun 12)
4	Projected ETA after disruption > required due date	Mark order at risk, calculate delay days, and move to mode-switch evaluation	Delay = MAX(ETA after disruption – due date, 0)
5	Priority Product A order is at risk	Compare ocean vs air vs split and choose option that keeps fulfillment ≥98%	Core optimization step
6	Delay days = 0	Keep ocean mode	No need to incur extra cost
7	Delay days > 0 for priority Product A order	Switch or split late-risk quantity to airfreight	Only expedite what is late (not full shipment)
8	Switched Product A quantity from ocean to air > 0 pcs	Show emergency airfreight cost and include in comparison	Air cost/pc = 4.50 × 1.2 = USD 5.40/pc
9	Airfreight cost ≤ 10% of consignment value	Prefer air transport to protect service level	Fast decision shortcut for high-value urgency
10	Container cost rises ≥20% OR FCL utilization < 70%	Re-optimize LCL/FCL/split and consolidate shipments	Avoid inefficient container usage post-disruption
11	Projected DC stock < 1 week demand	Trigger red alert, reserve inventory, expedite replenishment, or reallocate from another DC	Directly from safety buffer rule
12	Planned DC/distributor stock > 85% capacity	Split replenishment, pull forward fewer units, or reroute overflow to alternate DC/partner	Prevent congestion/capacity overflow
13	Distributor fulfillment rate < 95% (rolling 4 weeks)	Freeze new allocations, activate alternate distributor or MK direct support, and escalate SLA breach	Protect service reliability

DISRUPTION SCENARIO

15 days

LT shock

20%

Container cost spike

114,483

Product A in scope

74,493

Qty. late if ocean

113,738

Prodetected qty

99.34%

Coverage by due date

\$408,707

Recovery cost

\$3,806

Savings vs full air

Shipment	SKU A Qty	Before ETA	After ETA (+15d)	Delay Days	After Ocean Cost	Air Cost	Recommendation	Recommended Cost
#3 DC-US-W	20,000	Jun 2006	Jun 2021	0 (ok)	\$4,971	\$108,000	OCEAN ✓	\$4,971
#6 DC-US-W	20,000	Jun 2012	Jul 2003	0 (ok)	\$5,280	\$108,000	OCEAN ✓	\$5,280
#8 DC-US-W	11,000	Jun 2019	Jul 2015	1	\$4,289	\$59,400	SPLIT 98/2	\$58,298
#11 DC-US-W	13,000	Jun 2026	Aug 2007	8	\$5,074	\$70,200	SPLIT 98/2	\$68,897
#13 DC-US-W	13,773	Jun 2026	Aug 2019	15	\$4,294	\$74,374	SPLIT 98/2	\$72,973
#15 DC-US-W	7,058	Jul 2017	Aug 2031	27	\$2,423	\$38,113	AIR 100% ⚠	\$38,113
#18 DC-US-W	2,040	Jul 2024	Sep 2007	29	\$1,176	\$11,016	AIR 100% ⚠	\$11,016
#20 DC-US-W	3,541	Aug 2006	Sep 2020	36	\$1,727	\$19,121	AIR 100% ⚠	\$19,121
#23 DC-US-W	12,041	Aug 2008	Sep 2022	43	\$4,814	\$65,021	AIR 100% ⚠	\$65,021
#26 DC-US-W	12,040	Aug 2015	Sep 2029	50	\$4,766	\$65,016	AIR 100% ⚠	\$65,016

LIVE DASHBOARD

DC Stock vs Safety Buffer

40,000

units by due date

Safety Buffer: 112,203 units

Distributor Fulfillment Rate

99.34%

projected w/ split/air recovery plan

Target: ≥98.0%

Max Shipment Delay Days

+50

days on at-risk shipments

Target: 0 delay day

Priority	Action	Trigger	Timeline	Owner
P1 — IMMEDIATE	Switch 5 US Product A lines to 100% Air	Delay > 27 days → Air cost 5.1% of value	Today	MK Logistics
P1 — IMMEDIATE	Apply 98/2 split on 3 borderline US lines	Delay 1–15 days → protect 98% fulfillment	Today	MK Logistics
P2 — 48 HOURS	Activate DC safety buffer stock hold	Stock < 1-wk demand at DC-US-W	48 hrs	DC Operations
P3 — 7 DAYS	Re-evaluate container bookings (+20% cost)	Ocean cost shock → re-optimize LCL/FCL	7 days	Procurement

AI VALUE STATEMENT

Where AI adds most value

AI adds the most value in optimising the mode-switch decision, not in simply predicting the delay. Delay prediction is the early-warning trigger; the value is created when AI converts that warning into a shipment-level action: keep ocean, switch to 100% air, or use a 98/2 split by due date, launch priority, service target and cost-to-value threshold.

Apply to the model

The scenario creates a +15-day ocean lead-time shock and +20% ocean freight cost. Based on the table above, the system should keep 2 Product A lines on ocean, use a 98/2 split for 3 lines that become late but were originally recoverable, and use 100% airfreight for 5 lines already late before the disruption.

Priority-case decision

For the 20,000-unit US East launch due 15-May-2026, AI should recommend 100% airfreight because disrupted ocean misses launch by 15 days. For the 13,773-unit UK priority order due 12-Jun-2026, AI should recommend a 98/2 split if the $\geq 98\%$ target is acceptable; use full air only if the customer requires 100% launch completion.

Data AI needs

AI needs live port congestion signals, vessel ETD/ETA updates, port dwell/queue time, carrier reliability, lane lead-time history, ocean and air rates, air capacity, DC inventory vs safety buffer, distributor demand and due dates, Product A quantity/CBM/value, launch-priority flags, and business rules such as 98% fulfilment and the 10% air-cost/value threshold.

Business outcome improved

The business outcome improved is service-cost balance. In the current output, the recommended plan protects 113,738 Product A units by due date and delivers 99.34% fulfilment. It costs about \$369.9k more than disrupted ocean-only, but avoids the customer-service failure shown by red-alert KPIs and saves about \$3.8k versus full air on the at-risk quantity.

Operational recommendation

Use AI as a disruption control tower: predict risk early, then re-optimize the mode mix daily as ETAs, rates, capacity and stock positions change. Red alerts should automatically trigger a re-run whenever Shipment Delay Days > 0 or DC Stock Level falls below the safety buffer.

Strongest AI value

predictive delay detection plus cost/service optimisation, with the mode-switch optimiser being the decision engine that produces the measurable business benefit.

Appendix A: Key Rate Card & Assumptions

APPENDIX

Freight Rates (from case PDF)

Mode	US Rate	UK Rate	AU Rate	Capacity (CBM)	Lead Time
FCL 40ft	\$5,200	\$4,200	\$2,000	65 CBM	US:50d / UK:40d / AU:25d
FCL 20ft	\$3,000	\$2,500	\$1,100	28 CBM	US:50d / UK:40d / AU:25d
LCL	\$200/CBM	\$70/CBM	\$35/CBM	N/A	US:55d / UK:45d / AU:30d
Air (ref)	~\$5.40/pc	~\$5.40/pc	~\$5.40/pc	N/A	US:6-14d / UK:5-7d / AU:5-8d

Monthly Freight Cost Summary

Month	US SKUs (A,B,F)	UK SKUs (C,D)	AU SKU (E)	Total
May	\$12,442	\$1,081	\$162	\$13,685
June	\$12,121	\$4,638	\$66	\$16,825
July	\$13,155	\$5,640	\$176	\$18,971
TOTAL	\$37,718	\$11,359	\$404	\$49,481

Key Assumptions

- 1. All US SKUs (A,B,F) consolidated to DC-US-W (lower \$18/CBM handling vs \$20/CBM at DC-US-E; same 50-day ocean LT)
- 2. UK priority order of 13,773 pcs SKU A treated as UK launch order per case brief (even though SKU A is listed as US market in product master)
- 3. Base pipeline uses backward scheduling from customer due date. Shipments with ETD before May 15 assumed to already be in transit
- 4. Air cost: \$4.50/pc base × 1.20 disruption multiplier = \$5.40/pc. Applied only to Product A (high-value, time-sensitive)
- 5. Distribution burden = total distribution cost / ex-works value. AU 5.74% = high because SKU E ex-works is only \$30/unit
- 6. Distributor contingency switching costs based on FMCG/toy industry standard: 5% of at-risk revenue + onboarding fees